

Business Requirements Document (BRD)

Customer Segmentation & Sales Intelligence Platform using Machine Learning and Power BI

Version: 1.0 (Initial Draft)

Prepared By: Rizwan Hussain

Project Type: End-to-End Data Analytics & Machine Learning Project

Industry: Retail / E-commerce

Status: Initial Business Requirements

Date: July 2026

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Project Overview

This project aims to develop an end-to-end customer segmentation solution for Zandra Retail Ltd. using SQL, Python, Machine Learning, and Power BI. The solution will analyze historical transaction data, identify customer segments through RFM analysis and K-Means clustering, and provide interactive dashboards to support strategic business decisions.

1. Introduction

Zandra Retail Ltd. is a UK-based online retail company that sells a wide range of products, including home décor, gifts, stationery, seasonal items, and lifestyle accessories to customers across multiple countries. As the business has grown, the company has accumulated a large volume of transaction data but lacks a structured approach to understanding customer purchasing behavior.

The purpose of this project is to analyze customer transaction data, identify meaningful customer segments using machine learning techniques, and develop an interactive Power BI dashboard that supports strategic decision-making. The project aims to enable the business to improve marketing effectiveness, customer retention, and overall sales performance through data-driven insights.

2. Business Background

The company currently sends promotional campaigns to all customers without considering their purchasing behavior or value to the business. This approach results in inefficient marketing spending and missed opportunities for personalized customer engagement.

Management requires a solution that can identify valuable customer groups, understand purchasing patterns, and provide actionable insights through interactive dashboards. Customer segmentation will enable the organization to target the right customers with appropriate marketing strategies and improve customer satisfaction.

3. Problem Statement

The organization currently lacks an effective method for identifying different customer groups based on purchasing behavior. As a result, marketing campaigns are generalized, customer retention efforts are limited, and management has insufficient visibility into customer value and purchasing trends.

Without customer segmentation, the company cannot effectively prioritize marketing resources, identify loyal customers, or develop targeted retention strategies.

4. Business Objectives

The primary objectives of this project are:

- Analyze customer purchasing behavior using historical transaction data.
 - Segment customers into meaningful business groups.
 - Identify high-value, loyal, regular, and inactive customers.
 - Support personalized marketing campaigns.
 - Improve customer retention.
 - Increase marketing return on investment.
 - Develop an executive Power BI dashboard for business monitoring.
 - Provide actionable recommendations for business decision-makers.
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5. Project Scope

In Scope

- Business understanding
- Data understanding
- Data cleaning and preprocessing
- SQL business analysis
- Exploratory Data Analysis (EDA)
- Feature engineering
- RFM Analysis
- K-Means clustering
- Customer segmentation
- Cluster interpretation
- Power BI dashboard development
- DAX measures
- Power Query transformations
- Business recommendations

- Dashboard performance optimization
- Project documentation

Out of Scope

- Inventory optimization
- Demand forecasting
- Dynamic pricing analysis
- Recommendation systems
- Real-time dashboards
- Customer churn prediction
- Deployment of machine learning models into production

6. Stakeholders

Stakeholder	Responsibility	Expected Outcome
Chief Executive Officer (CEO)	Business Strategy	High-level business insights
Marketing Manager	Marketing Campaigns	Customer segmentation for targeted campaigns
Sales Manager	Sales Performance	Revenue and customer purchasing insights
Finance Team	Financial Monitoring	Revenue and customer profitability analysis
Data Analyst	Data Preparation & Analysis	Data cleaning, analysis, machine learning, and dashboard development

7. Business Requirements

The proposed solution should satisfy the following business requirements:

BR-01

Identify different customer groups based on purchasing behavior.

BR-02

Measure customer value using Recency, Frequency, and Monetary (RFM) analysis.

BR-03

Identify high-value customers for loyalty programs.

BR-04

Identify customers requiring retention campaigns.

BR-05

Develop an interactive executive dashboard.

BR-06

Provide country-wise and product-wise sales analysis.

BR-07

Present business recommendations based on analytical findings.

BR-08

Ensure dashboards support interactive filtering and business exploration.

8. Assumptions

The following assumptions are made at the beginning of the project:

- The transaction dataset accurately represents historical customer purchases.
- Customer IDs uniquely identify customers.
- Invoice records are reliable for customer behavior analysis.
- The dataset contains sufficient information for customer segmentation.
- Historical purchasing behavior reflects general customer trends.

These assumptions may be validated or revised during the data understanding phase.

9. Constraints

Potential project constraints include:

- Missing customer identifiers.

- Cancelled invoices.
 - Product returns represented by negative quantities.
 - Missing product descriptions.
 - Limited customer demographic information.
 - No explicit customer satisfaction or marketing campaign data.
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10. Success Criteria

The project will be considered successful if it:

- Successfully identifies meaningful customer segments.
 - Produces actionable business insights.
 - Supports marketing decision-making.
 - Delivers a professional Power BI dashboard.
 - Provides clear recommendations for improving customer engagement and retention.
 - Demonstrates a complete end-to-end analytics workflow.
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11. Deliverables

The final project will include:

- Business Requirements Document (Final Version)
- SQL Analysis Scripts
- Python Jupyter Notebook
- Cleaned Dataset
- Feature Engineered Dataset
- Customer Segmentation Model
- Saved Machine Learning Model
- Power BI Dashboard (.pbix)
- Project Documentation
- GitHub Repository

- Business Recommendations Report
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12. Risks

Potential risks include:

- Poor data quality affecting model performance.
- Incomplete customer information.
- High number of cancelled transactions.
- Incorrect interpretation of customer behavior.
- Business assumptions differing from actual organizational practices.

These risks will be monitored and addressed throughout the project lifecycle.

13. Future Enhancements

The following enhancements are outside the scope of the current project but may be considered in future phases:

- Customer churn prediction.
 - Product recommendation engine.
 - Customer lifetime value prediction.
 - Sales forecasting.
 - Marketing campaign effectiveness analysis.
 - Real-time dashboard integration.
 - Automated data refresh using cloud services.
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